

Technische Universität Berlin
Fakultät VI - Planen Bauen Umwelt
Institut für Bauingenieurwesen
Fachgebiet Systemtechnik baulicher Anlagen

Multi-Physics approaches for modeling civil systems
Integrated Model Assignment

Submitted by: Group F

Iveli de Paula Sousa
Mohammed Amir Aziz Khoshnaw
Remyasree Chodon Puthiyoth
Taaliya Siddiqui
Yunsu Jang

Introduction

The Martin-Gropius-Bau, also known as Gropius Bau, is one of the most important exhibition buildings Europe (1). Located in Berlin, Germany, within the Kreuzberg neighborhood, Gropius Bau was originally a museum of applied arts and has been a listed historical monument since 1966. Yearly, it attracts over half a million visitors, with a peak of 700,000 in 2014.

Built in 1881, the Gropius Bau incorporates a solid stone construction with steel elements and an atrium with natural light. The presentation by the architects working on the project for a new low-tech approach provided important information. One of the current challenges for climate control in the chambers that house the museum's objects is humidity control. In addition to insulation, maintaining the ideal conditions of 20°C and 50% humidity is a constant problem. Currently, this challenge is mitigated by several individual dehumidifiers, which on the other hand detracts from the immersive artistic experience. In addition, so far 2% of electricity is produced by solar panels and the administration is open to the idea of installing more.

The goals of this work are two: first, to explore the energy production by **installing additional solar panels**, and second, to create a solution that **replaces the different dehumidification units with an integrated ventilation system**, guaranteeing temperature stability. For both scenarios, there is a comparison of energy required to keep the building maintenance. By addressing these objectives, we aim to preserve the artistic ambience of the Gropius Bau while at the same time adopting sustainable energy practices and optimizing environmental control.

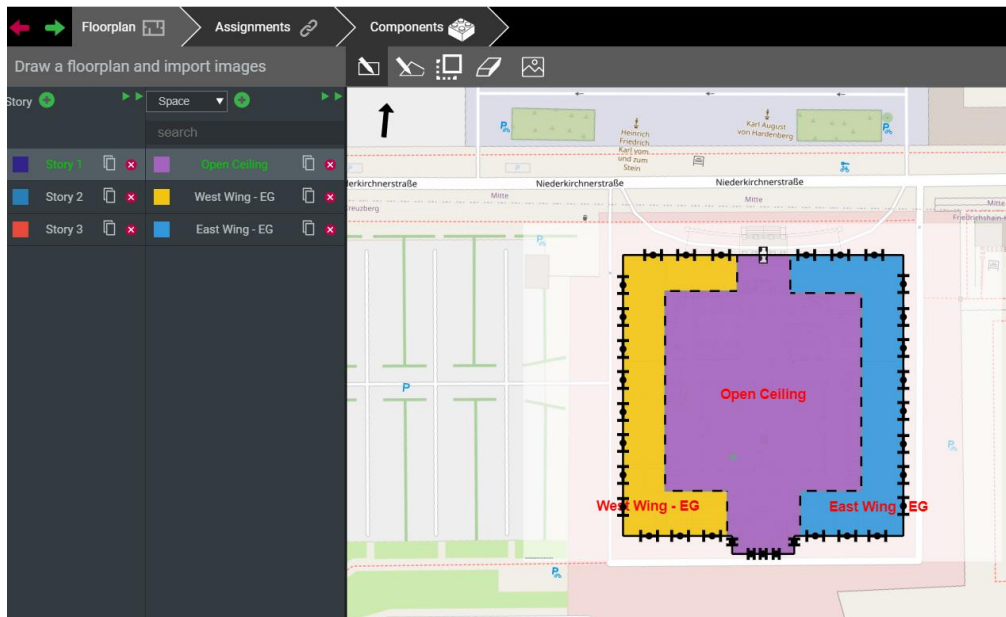
Methodology

Simulation Tools Used:

In our assignment, we used mostly three key simulation tools: FloorspaceJS, OpenStudio, and EnergyPlus.

- **FloorspaceJS:**

FloorspaceJS was used to create a simplified representation of the spatial geometry of the Gropius Bau building. This involves assigning properties such as construction sets and thermal zones to the building's spaces. Initially, an attempt was made to build the geometry using SketchUp, importing and converting the IFC files provided by the architects. However, due to computational constraints, this approach was very unsuccessful. A complex geometry increased the running time of the simulation too much in both OpenStudio and EnergyPlus, as well as generating several crashes in both OpenStudio and EnergyPlus during the model creation.



- **OpenStudio:**

OpenStudio served as a cross-platform, open-source software development kit (SDK) for building energy modeling and analysis. It provided a comprehensive framework for creating and analyzing building energy models, enabling various assessments such as energy efficiency improvements, code compliance, and sustainability evaluations. (3)

- **EnergyPlus:**

EnergyPlus, a whole building energy simulation program, was used to model the energy consumption of the Gropius Bau building. This included heating, cooling, ventilation, lighting, plug loads, and process loads, as well as water use. The simulation inputs were intertwined with those of OpenStudio, facilitating a holistic approach to whole building energy modeling. (4)

Simulation Setup:

- To initiate the simulations, the Parametric Analysis Tool from OpenStudio was installed by cloning the OpenStudio GitHub repository (5).
- Weather data for the simulations were obtained from the EnergyPlus website. Specifically, we used an EPW (EnergyPlus Weather) file sourced from the designated weather location for Berlin, Germany. Additionally, design day files (DDY) from EnergyPlus were imported to ensure the whole weather data coverage.
- For the simulations conducted in OpenStudio, a climate zone classification of 5A was utilized, according to the ANSI/ASHRAE 169-2020 world climate zones map. (6)

Modelling the Martin-Gropius-Bau Building

After creating the building geometry, it was imported into both simulation programs for further analysis.

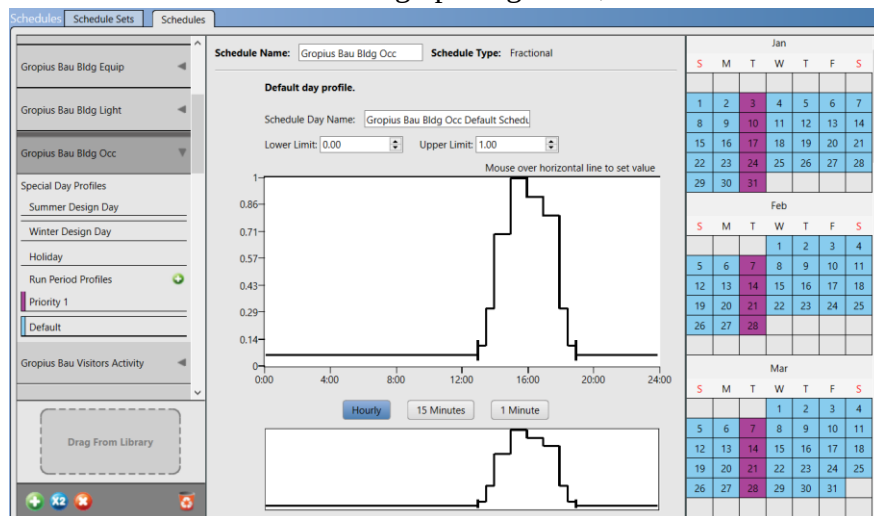
Schedule:

The schedule feature was used to define time-dependent profiles for various parameters such as occupancy, lighting, equipment usage, temperature setpoints, and airflow rate. For this assignment, each one of the parameters were adjusted to accurately represent the patterns of the Gropius Bau. For instance, the lighting schedule was adjusted to reduce

illumination during non-open hours while maintaining a minimum level for security staff working on the building 24/7.

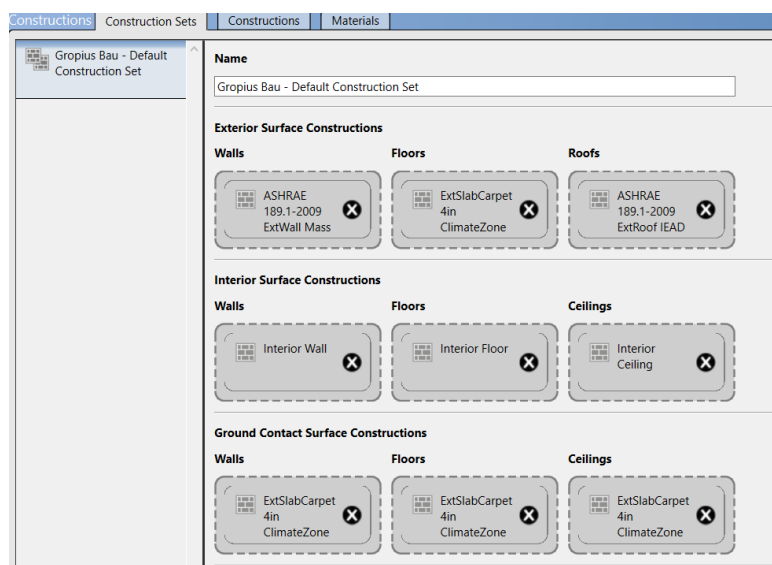
The occupancy schedule was set similarly, with a maximum capacity at night of only 10% due to the presence of security staff. One hour before and one hour after opening time, the occupancy slightly increases thanks to its employees - exhibition supervision personal, cashiers, guides and so on. The occupancy is obviously the highest during the opening time – from 14:00 to 18:00, reaching 100% (upper limit 1) at 15:00. On the figure below, the calendar highlight in pink Tuesdays – this is a priority created to reflect that the museum is closed.

The same logic was applied to other schedules, such as electrical equipment usage, which peaks the use of dehumidifiers during opening times, and infiltration.



Construction Sets:

Then we begin assign construction sets for each of the building elements (walls, roofs, floors, windows, etc.) within each thermal zone. This part defines the materials and properties of the building elements, such as insulation thickness, conductivity, and solar heat gain coefficients.



Specify Internal Loads and Occupancy:

Due to its complexity a standard HVAC system was assigned with an specific temperature standpoint.

Loads

Name: 189.1-2009 - Lobby - CZ4-8 Lights Definition

Lighting Power: W **Watts Per Space Floor Area:** 12.593775 W/m² **Watts Per Person:** W/person

Fraction Radiant: 0.000000 **Fraction Visible:** 0.000000

Return Air Fraction: 0.000000

People Definitions

Lights Definitions

- 189.1-2009 - Lobby - CZ4-8 Lights Definition
- 189.1-2009 - WholeBuilding - Lg Office - CZ4-8 Lights
- Lights Definition 1

Luminaire Definitions

Electric Equipment

Spaces:

For simplification and to decrease the computational effort, there are only two space zones with different parameters. The Foyer space allows natural ventilation and has a lower minimum temperature allowed. Inside the Exhibition Space, there is no natural ventilation, the temperature is to be kept at 20°C and humidity level at a max of 50%.

Filter: Load Type

Show all loads

Space Type Name	All	Rendering Color	Default Construction Set	Default Schedule Set
	☐		Apply to Selected	Apply to Selected
Exhibition Space	☐		Gropius Bau - Default Con:	GropiusBau - Schedule Set
Foyer	☐		Gropius Bau - Default Con:	GropiusBau - Schedule Set

Simulations

EnergyPlus Simulation

The initial simulation focused on assessing the electricity-saving potential achievable by replacing individual dehumidifiers in exhibition rooms with an HVAC system featuring an automated humidity control. To conduct this simulation, we adjusted the HVAC parameters and multiplied the dehumidifiers by a factor of zero, so it would be deactivated throughout the year.

Initial Condition:

Below is the screenshot from museum's existing setup report with individual dehumidifiers distributed across exhibition rooms.

Program Version:EnergyPlus, Version 23.2.0-7636e6b3e9, YMD=2024.02.13 17:19

Tabular Output Report in Format: HTML

Building: Gropius Bau

Environment: RUN PERIOD 1 ** BERLIN - DEU IWEC Data WMO#=103840

Simulation Timestamp: 2024-02-13 17:19:13

Report: Annual Building Utility Performance Summary

For: Entire Facility

Timestamp: 2024-02-13 17:19:13

Values gathered over 8760.00 hours

Site and Source Energy

	Total Energy [GJ]	Energy Per Total Building Area [MJ/m2]	Energy Per Conditioned Building Area [MJ/m2]
Total Site Energy	4706.04	314.58	314.58
Net Site Energy	4706.04	314.58	314.58
Total Source Energy	15456.90	1033.23	1033.23
Net Source Energy	15456.90	1033.23	1033.23

Simulation Results:

And below are the simulation results of the new implementation.

Program Version:EnergyPlus, Version 23.2.0-7636e6b3e9, YMD=2024.02.13 17:25

Tabular Output Report in Format: HTML

Building: Gropius Bau

Environment: RUN PERIOD 1 ** BERLIN - DEU IWEC Data WMO#=103840

Simulation Timestamp: 2024-02-13 17:25:22

Report: Annual Building Utility Performance Summary

For: Entire Facility

Timestamp: 2024-02-13 17:25:22

Values gathered over 8760.00 hours

Site and Source Energy

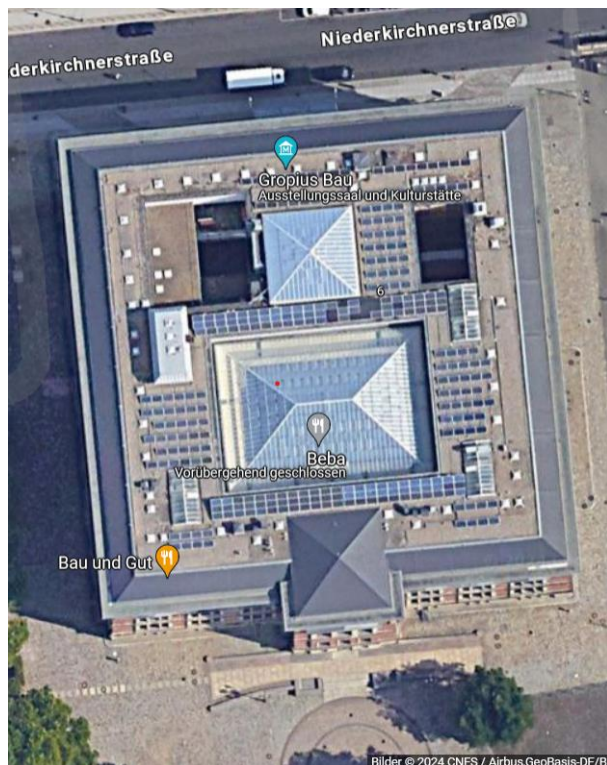
	Total Energy [GJ]	Energy Per Total Building Area [MJ/m2]	Energy Per Conditioned Building Area [MJ/m2]
Total Site Energy	4548.67	304.06	304.06
Net Site Energy	4548.67	304.06	304.06
Total Source Energy	14998.06	1002.56	1002.56
Net Source Energy	14998.06	1002.56	1002.56

Upon implementing the new HVAC system with humidity control, the total energy consumption decreased from 4706.04 GJ to 4548.67 GJ annually, which represents a 3.35% reduction in energy consumption.

OpenStudio Simulation

Using OpenStudio in conjunction with Radiance, a light condition simulation software (6), the potential energy generation from solar panels could be simulated.

Looking at recent satellite image of the museum (2024), almost all the roof has already installed solar panels. The area on the upper side of the image could be used to install 18 new solar panels with an angle of 36° facing south.



Below are the simulation results on initial conditions:

OpenStudio Results

Model Summary

Building Summary	
Data	Value
Building Name	Gropius Bau
Total Site Energy	4,311,307 kBtu
Total Building Area	161,026 ft ²
Total Site EUI	26.77 kBtu/ft ²
OpenStudio Standards Building Type	n/a

Figure 1

And those are the results by installing new solar panels:

OpenStudio Results

Model Summary

Building Summary	
Data	Value
Building Name	Gropius Bau
Total Site Energy	4,298,028 kBtu
Total Building Area	161,026 ft^2
Total Site EUI	26.69 kBtu/ft^2
OpenStudio Standards Building Type	n/a

Figure 2

This shows that installing 18 new solar panels on the upper portion of the roof at an angle of 36° facing south resulted in a reduction of total yearly energy consumption from 4311307 kBtu to 4298026 kBtu. But unfortunately, the saving potential per year would be only 0,31%.

Both the reports from EnergyPlus and from OpenStudio, which are inside the attachments folder, can provide more detailed information and insights about both the initial models and the simulations.

Discussion

Obviously, the simulation has some limitations. One of them is the geometrical model, which doesn't accurately represent the real building, but has been adapted in this way mainly for computational reasons. Regarding the second part of the simulation, although the potential energy savings are quite small, it showed how Gropius Bau has already done an excellent job by installing solar panels on the entire roof. The only part of the roof where solar panels have not been installed is exactly on the side that receives the least sunlight (northwest).

On the other hand, it might be interesting to install a new HVAC capable of removing the humidity of the exhibition rooms. The initial financial investment and the increase on the electricity consumption of this new HVAC system can outweigh the energy potential of eliminating the individual dehumidifiers – while not distracting the visitors from the art.

An interesting information that might be useful for future exploration/research is that the German government, in the face of climate change, has a new guideline regarding the ideal temperature for museums. The degrees of the new temperature requirements are between 15 and 26 °C and a relative humidity of between 40% and 60% (7). If it is guaranteed that museum objects will not be damaged, there is certainly great potential for energy savings.

References

- (1) *Gropius Bau*. Museumsportal Berlin. 09.02.2023, [online]
<https://www.museumsportal-berlin.de/de/museen/martin-gropius-bau/>
- (2) *OpenStudio*. (2023). [Software]
<https://openstudio.net/>
- (3) U.S. Department of Energy. (2021) *EnergyPlus* (V9.6.0) [Software].
<https://energyplus.net/downloads/>
- (4) Nrel. (n.d.). *GitHub - NREL/OpenStudio-PAT: The Parametric Analysis Tool (PAT) is part of the OpenStudio collection of software tools*. GitHub.
<https://github.com/NREL/OpenStudio-PAT>
- (5) Norouziasas, A., Tabadkani, A., Rahif, R., Amer, M., Van Dijk, D., Lamy, H., & Attia, S. (2023). Implementation of ISO/DIS 52016-3 for adaptive façades: A case study of an office building. *Building and Environment*, 235, 110195. DOI: [10.1016/j.buildenv.2023.110195](https://doi.org/10.1016/j.buildenv.2023.110195)
- (6) *Radiance* — *Radsite*. (2019). [Software].
<https://www.radiance-online.org/>
- (7) *Neue Standards für die Museumsklimatisierung angesichts der Klimakrise* | *Deutscher Museumsbund e.V.* (n.d.). Deutscher Museumsbund e.V.
<https://www.museumsbund.de/museumsklimatisierung/>

Appendices:

- ***floorplan.json*** – File with geometry and space configuration used on this simulation
- ***Berlin Weather Data.zip*** – EnergyPlus weather data
- **EnergyPlus Results** – folder with both reports from EnergyPlus Simulation
- **OpenFoam Results** – folder with both reports from OpenFoam Simulation